

Dwell-Conditioned Cycle-to-Cycle Switching-Time Feedback for Mode-Dependent Switched-Affine Systems

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Abstract

A static cycle-to-cycle switching-time law can regulate a periodic trajectory at a fixed cycle period, but its unconstrained output may produce negative or otherwise infeasible dwell durations. We prevent this by scaling the full vector of raw switching-instant offsets by the largest factor in $[0, 1]$ that satisfies every minimum-dwell constraint. The scaling preserves the action direction and cycle period. For a schedule with N intervals, it requires an $O(N)$ scan and no online solver. In the linearized cycle model, the resulting matrix belongs to the segment $A(\beta) = (1 - \beta)\Phi + \beta A_{cl}$. A common quadratic Lyapunov function for the two endpoints proves contraction for any sequence of conditioning factors. We also derive a fixed-period one-cycle model by exact augmented matrix-exponential propagation for a prescribed sequence of affine modes, allowing both the state matrices and affine vectors to vary. The derivation recovers the known switching sensitivity and reduces to the switched-actuator model as a special case. On a nine-interval converter benchmark, the analytical Jacobians match central differences with relative errors of $8.518\text{e-}12$ and $8.344\text{e-}11$, and the linearization residual has slope 2.000. For a large initial error, an aggressive raw law requests a $-43.045\text{ }\mu\text{s}$ dwell. Conditioning enforces the assumed $3\text{ }\mu\text{s}$ bound and settles faster than a feasible conservative law in exact nonlinear simulation. The Lyapunov result applies to the linearized model, and the nonlinear stability result is local.

Keywords: switched-affine systems, periodic trajectories, switching-time control, dwell-time conditioning, one-cycle map, DC-DC converter

1 Introduction

Power converters regulate voltage, current, and energy flow by changing semiconductor switch configurations. Their controllers must respect fixed switching frequency, minimum dwell duration, current limits, voltage-balance requirements, and admissible switch commands within short sampling intervals. Predictive methods handle such constraints directly, but their online cost remains an implementation concern. Recent designs reduce

this cost through lower-dimensional parameterizations Saeed et al. (2022), custom fixed-frequency solvers Yang et al. (2022), and candidate pruning or approximation Augustine and Patil (2023); Sakha and Kamalapurkar (2025). Static feedback avoids this online optimization, but its requested switching action may still violate the physical timing constraints.

This paper considers a narrower control setting. A prescribed sequence of affine modes repeats with fixed period, and the controller moves only the interior switching instants from one cycle to

the next. The target is therefore a periodic trajectory rather than an equilibrium. This setting differs from methods that choose the active mode as a function of the state [Egidio et al. \(2022\)](#); [Russo et al. \(2026\)](#) and from methods that optimize a complete switching schedule online. Here, the cycle-start error produces a static correction to the next cycle's switching instants while preserving the mode order and cycle boundaries.

The underlying cycle model follows established results. Patiño, Riedinger, and Ruiz derived switching-instant sensitivities for cyclic systems with mode-dependent dynamics and used them in predictive control [Patiño et al. \(2010\)](#). Marcolino, Galvão, and Kienitz derived a fixed-period one-cycle model, a discrete LQR timing law, and dwell-constrained predictive control for an LTI plant driven by switched actuator levels [Marcolino et al. \(2021\)](#). In their standalone LQR comparison, sequential saturation can move the final switching instant; in the reported example, the resulting response diverges [Marcolino et al. \(2021\)](#). More generally, unconstrained timing corrections can violate coupled dwell limits, while clipping individual offsets can alter the requested correction direction. Thus, replacing online optimization with static feedback does not by itself provide an admissible switching schedule.

We address this implementation problem without redesigning the raw controller. Given a raw vector of switching-instant offsets, the conditioner applies the largest scalar multiple that satisfies every minimum-dwell constraint. Scaling the complete vector preserves its direction, the prescribed mode order, and both cycle boundaries. The feasible factor is obtained by scanning the N dwell changes, which gives $O(N)$ online cost and requires no optimization solver. In the linearized cycle model, the conditioned closed-loop matrix lies on the segment $A(\beta) = (1 - \beta)\Phi + \beta A_{cl}$. A common quadratic Lyapunov function for the endpoints certifies contraction for any sequence of conditioning factors. These are the paper's two main contributions. To support them, we derive the fixed-period one-cycle model for mode-dependent affine dynamics, recover the published switching-sensitivity structure, verify the Jacobians numerically, and evaluate the controller in exact nonlinear simulation of a nine-interval converter benchmark. The LQR laws are comparison controllers rather than new synthesis methods. The nonlinear stability

guarantee remains local, and the converter study does not constitute hardware validation.

2 Fixed-period one-cycle model

2.1 Switched-affine cycle and timing coordinates

Consider

$$\dot{\mathbf{x}}(t) = A_{\sigma(t)}\mathbf{x}(t) + \mathbf{b}_{\sigma(t)}, \quad \mathbf{x}(t) \in \mathbb{R}^n, \quad (1)$$

where each cycle follows the fixed mode sequence $\Omega = (\omega_1, \dots, \omega_N)$ with nominal boundaries

$$0 = \bar{t}_0 < \bar{t}_1 < \dots < \bar{t}_N = T, \quad \bar{d}_i = \bar{t}_i - \bar{t}_{i-1}. \quad (2)$$

The nominal schedule defines a reference periodic trajectory $\bar{\mathbf{x}}(t)$. We use its cycle anchor $\mathbf{x}^* = \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}(T)$ to define the cycle-start error $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$.

Only the interior switching instants move:

$$t_{k,i} = \bar{t}_i + \delta\tau_{k,i}, \quad i = 1, \dots, N-1, \quad (3)$$

$$\delta\tau_{k,0} = \delta\tau_{k,N} = 0.$$

The vector $\delta\tau_k$ contains the $N-1$ independent offsets. The corresponding N dwell changes satisfy

$$\delta\mathbf{d}_k = D\delta\tau_k, \quad D_{ij} = \begin{cases} 1, & i = j, \\ -1, & i = j+1, \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

for $i = 1, \dots, N$ and $j = 1, \dots, N-1$. Thus $\delta d_{k,i} = \delta\tau_{k,i} - \delta\tau_{k,i-1}$. Because $\mathbf{1}^\top D = 0$, every offset vector preserves the cycle period T . We denote the exact one-cycle map by $\mathbf{x}_{k+1} = \mathcal{P}(\mathbf{x}_k, \delta\tau_k)$.

2.2 Exact augmented propagation

Define the augmented state, the projection onto the physical state, and the mode generators as

$$\mathcal{X} = \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}, \quad \Pi = [I_n \ 0], \quad F_i = \begin{bmatrix} A_{\omega_i} & \mathbf{b}_{\omega_i} \\ \mathbf{0}^\top & 0 \end{bmatrix}. \quad (5)$$

The affine dynamics on interval i then become $\dot{\mathcal{X}} = F_i \mathcal{X}$. For an interval of duration s , let $E_i(s) = e^{F_i s}$ and $\phi_i = E_i(\bar{d}_i)$. Since $d_{k,i} = \bar{d}_i + \delta d_{k,i}$, exact

propagation over one cycle gives

$$\mathcal{X}_{k+1} = E_N(\bar{d}_N + \delta d_{k,N}) \cdots E_1(\bar{d}_1 + \delta d_{k,1}) \mathcal{X}_k, \quad (6)$$

The rightmost factor propagates the first interval. Let

$$\bar{\mathcal{X}}_i = \phi_i \cdots \phi_1 \bar{\mathcal{X}}_0, \quad \tilde{\mathbf{e}}_k = \begin{bmatrix} \mathbf{e}_k \\ 0 \end{bmatrix}. \quad (7)$$

Periodicity implies $\phi_N \cdots \phi_1 \bar{\mathcal{X}}_0 = \bar{\mathcal{X}}_0$. Subtracting the nominal cycle from (6) gives the exact augmented terminal error

$$\begin{aligned} \tilde{\mathbf{e}}_{k+1} &= E_N(\bar{d}_N + \delta d_{k,N}) \cdots E_1(\bar{d}_1 + \delta d_{k,1}) \\ &\quad \times (\bar{\mathcal{X}}_0 + \tilde{\mathbf{e}}_k) - \phi_N \cdots \phi_1 \bar{\mathcal{X}}_0. \end{aligned} \quad (8)$$

2.3 First-order expansion in dwell durations

Because F_i is constant on interval i , differentiation with respect to the interval duration gives

$$\frac{d}{ds} e^{F_i s} = F_i e^{F_i s}, \quad (9)$$

and each perturbed transition satisfies

$$E_i(\bar{d}_i + \delta d_{k,i}) = \phi_i + F_i \phi_i \delta d_{k,i} + \mathcal{O}(\delta d_{k,i}^2). \quad (10)$$

Substitute (10) into the ordered product and retain terms with at most one dwell perturbation. With $E_i = E_i(\bar{d}_i + \delta d_{k,i})$ on the left, this gives

$$\begin{aligned} E_N \cdots E_1 &= \tilde{\Phi} + \sum_{i=1}^N \phi_N \cdots \phi_{i+1} F_i \phi_i \cdots \phi_1 \delta d_{k,i} \\ &\quad + \mathcal{O}(\|\delta \mathbf{d}_k\|^2), \quad \tilde{\Phi} = \phi_N \cdots \phi_1, \end{aligned} \quad (11)$$

where an empty product equals the identity. Inserting this expansion into (8) cancels the nominal term. The products $\delta d_{k,i} \tilde{\mathbf{e}}_k$ are second order and therefore omitted. Using $\phi_i \cdots \phi_1 \bar{\mathcal{X}}_0 = \bar{\mathcal{X}}_i$, the remaining first-order terms are

$$\tilde{\mathbf{e}}_{k+1} = \tilde{\Phi} \tilde{\mathbf{e}}_k + \sum_{i=1}^N \tilde{\mathbf{g}}_i \delta d_{k,i} + \mathcal{O}(\|[\mathbf{e}_k^\top \delta \mathbf{d}_k^\top]\|^2), \quad (12)$$

with

$$\tilde{\mathbf{g}}_i = \phi_N \cdots \phi_{i+1} F_i \bar{\mathcal{X}}_i. \quad (13)$$

Projecting onto the physical state gives

$$\mathbf{e}_{k+1} = \Phi \mathbf{e}_k + G_d \delta \mathbf{d}_k + \mathcal{O}(\|[\mathbf{e}_k^\top \delta \mathbf{d}_k^\top]\|^2), \quad (14)$$

where

$$\Phi = e^{A_{\omega_N} \bar{d}_N} \cdots e^{A_{\omega_1} \bar{d}_1}, \quad G_d = \Pi[\tilde{\mathbf{g}}_1 \cdots \tilde{\mathbf{g}}_N]. \quad (15)$$

2.4 Switching-instant offsets

Substituting $\delta \mathbf{d}_k = D \delta \boldsymbol{\tau}_k$ into (14) gives the model in the independent timing coordinates

$$\begin{aligned} \mathbf{e}_{k+1} &= \Phi \mathbf{e}_k + \Gamma_\tau \delta \boldsymbol{\tau}_k + \mathcal{O}(\|[\mathbf{e}_k^\top \delta \boldsymbol{\tau}_k^\top]\|^2), \\ \Gamma_\tau &= G_d D. \end{aligned} \quad (16)$$

Column i of D increases dwell i and decreases dwell $i+1$. Therefore $\Gamma_{\tau,i} = \Pi(\tilde{\mathbf{g}}_i - \tilde{\mathbf{g}}_{i+1})$. Since F_{i+1} commutes with $\phi_{i+1} = e^{F_{i+1} \bar{d}_{i+1}}$, factoring the adjacent terms yields

$$\begin{aligned} \Gamma_{\tau,i} &= e^{A_{\omega_N} \bar{d}_N} \cdots e^{A_{\omega_{i+1}} \bar{d}_{i+1}} \\ &\quad \times [(A_{\omega_i} - A_{\omega_{i+1}}) \bar{\mathbf{x}}(\bar{t}_i) + \mathbf{b}_{\omega_i} - \mathbf{b}_{\omega_{i+1}}] \end{aligned} \quad (17)$$

The bracket is the difference between the adjacent affine vector fields at the switching state. Equation (17) has the switching-sensitivity structure reported in [Patiño et al. \(2010\)](#); here it follows from dwell perturbations in independent fixed-period timing coordinates.

If every interval instead shares $A_{\omega_i} = A_c$ and $\mathbf{b}_{\omega_i} = B_c \mathbf{u}_{i-1}$, then

$$\Phi = e^{A_c T}, \quad \Gamma_{\tau,i} = e^{A_c (T - \bar{t}_i)} B_c (\mathbf{u}_{i-1} - \mathbf{u}_i), \quad (18)$$

which matches Marcolino's switched-actuator timing model, apart from the sign convention for the input jump [Marcolino et al. \(2021\)](#). When A_{ω_i} depends on the mode, the jump also depends on the switching state, so this reduction does not apply in general.

3 Solver-free dwell-time conditioning

Let a static feedback law produce the raw switching-instant offset

$$\delta \boldsymbol{\tau}_k^{\text{raw}} = -K \mathbf{e}_k, \quad A_{\text{cl}} = \Phi - \Gamma_\tau K. \quad (19)$$

For a required minimum dwell duration $d_{\min}$, assume that the nominal schedule has the strict

margin

$$\mathbf{m} = \bar{\mathbf{d}} - d_{\min} \mathbf{1} > \mathbf{0}. \quad (20)$$

The raw action is feasible for errors in the polyhedron

$$\mathcal{C} = \{\mathbf{e} : \bar{\mathbf{d}} - D\mathbf{K}\mathbf{e} \geq d_{\min} \mathbf{1}\}. \quad (21)$$

For $\mathbf{e}_k \in \mathcal{C}$, the raw action needs no modification. Otherwise, define $\mathbf{q}_k = D\delta\boldsymbol{\tau}_k^{\text{raw}}$ and

$$\beta_k = \min \left\{ 1, \min_{i: q_{k,i} < 0} \frac{m_i}{-q_{k,i}} \right\}, \quad (22)$$

$$\delta\boldsymbol{\tau}_k^{\text{app}} = \beta_k \delta\boldsymbol{\tau}_k^{\text{raw}}.$$

where the minimum over an empty set is $+\infty$. We call β_k the conditioning factor.

Proposition 1 (Feasibility and maximality). *Under (20), (22) returns the largest $\beta \in [0, 1]$ such that every applied dwell duration satisfies $\bar{\mathbf{d}} + D(\beta\delta\boldsymbol{\tau}_k^{\text{raw}}) \geq d_{\min} \mathbf{1}$. The conditioned action preserves the cycle period and the direction of the raw action. Computing it takes $O(N)$ time and requires no online solver.*

Proof If $q_{k,i} \geq 0$, scaling cannot shorten dwell i . If $q_{k,i} < 0$, feasibility requires $\beta \leq m_i/(-q_{k,i})$. Intersecting these bounds with $[0, 1]$ gives (22) and proves maximality. The identity $\mathbf{1}^\top D = 0$ preserves T , and multiplication of the full action by one nonnegative scalar preserves its direction. A single pass over the N dwell changes evaluates all bounds. $\square$

In the linearized model, the applied action gives

$$\mathbf{e}_{k+1} = A(\beta_k)\mathbf{e}_k, \quad A(\beta) = (1 - \beta)\Phi + \beta A_{\text{cl}}. \quad (23)$$

Thus conditioning produces a matrix on the segment between the nominal matrix Φ and the raw closed-loop matrix A_{cl} , rather than an arbitrary saturated gain.

Proposition 2 (Common-Lyapunov contraction). *Let $P = P^\top > 0$, $\|\mathbf{x}\|_P = (\mathbf{x}^\top P \mathbf{x})^{1/2}$, and suppose*

$$\|\Phi\|_P \leq q_0 < 1, \quad \|A_{\text{cl}}\|_P \leq q_1 < 1. \quad (24)$$

Then every $A(\beta)$ with $\beta \in [0, 1]$ contracts in the same norm by a factor no greater than $q = \max(q_0, q_1) < 1$. The conditioned linearized model is therefore exponentially stable for every sequence $\{\beta_k\} \subset [0, 1]$, including the state-dependent sequences generated by (22).

Proof The induced-norm triangle inequality gives $\|A(\beta)\|_P \leq (1 - \beta)q_0 + \beta q_1 \leq q < 1$. Therefore $\|\mathbf{e}_k\|_P \leq q^k \|\mathbf{e}_0\|_P$ for any sequence. $\square$

When such a P exists, it can be computed offline from endpoint Lyapunov inequalities, which are standard semidefinite constraints [Boyd et al. \(1994\)](#). Online execution only evaluates the matrix-vector feedback law, $D\delta\boldsymbol{\tau}^{\text{raw}}$, and the scalar scan in (22).

The nonlinear result is local. Suppose that $\mathcal{P}$ is continuously differentiable while the mode order is preserved and that A_{cl} is Schur. The strict nominal margin and $\delta\boldsymbol{\tau}^{\text{raw}}(\mathbf{0}) = \mathbf{0}$ then place a neighborhood of the anchor inside $\mathcal{C}$. In that neighborhood, the conditioner is inactive, with $\beta = 1$, and the Jacobian of the exact closed-loop map is A_{cl} . The map therefore locally exponentially stabilizes the cycle anchor and its phase-locked reference periodic trajectory. [Proposition 2](#) covers arbitrary conditioning only for the linearized model and does not prove global stability of the nonlinear plant.

4 Converter case study

4.1 Three-cell converter benchmark and implementation

We use the three-cell multilevel DC-DC converter studied by [Patiño et al. \(2010\)](#), with state $\mathbf{x} = [v_{C_1}, v_{C_2}, i_L]^\top$. For binary switch commands $u_1, u_2, u_3 \in \{0, 1\}$,

$$A_{\mathbf{u}} = \begin{bmatrix} 0 & 0 & (u_2 - u_1)/C_1 \\ 0 & 0 & (u_3 - u_2)/C_2 \\ (u_1 - u_2)/L & (u_2 - u_3)/L & -R/L \end{bmatrix},$$

$$\mathbf{b}_{\mathbf{u}} = \begin{bmatrix} 0 \\ 0 \\ Eu_3/L \end{bmatrix}. \quad (25)$$

Both $A_{\mathbf{u}}$ and $\mathbf{b}_{\mathbf{u}}$ depend on the switch state. The plant therefore does not satisfy the common-state-matrix assumption in [\(18\)](#).

The reported physical mode sequence is $(0, 1, 3, 7, 2, 0, 4, 7, 4)$. The rounded cycle boundaries are $(0, 66, 88, 110, 132, 154, 220, 242, 264, 286)$ μs , and the reported cycle anchor is $\mathbf{x}^* = (9.9247, 19.2928, 0.9823)^\top$. Exact propagation with these rounded data does not return exactly to the anchor. We therefore reconcile the dwell vector

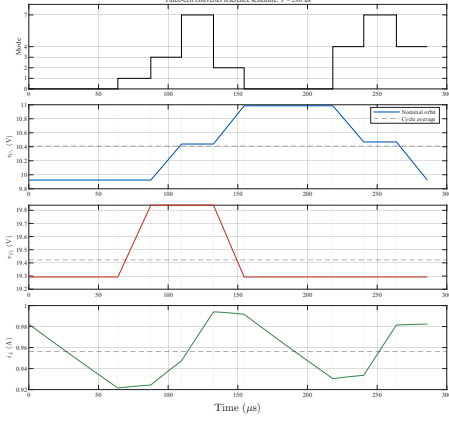


Figure 1 Reconciled reference schedule and exact periodic trajectory for the three-cell converter. Dashed horizontal lines show cycle averages. Dotted vertical lines mark the eight movable interior switching instants.

once by least squares, subject to the published $286 \mu\text{s}$ period, anchor, and $22 \mu\text{s}$ nominal-design bound. The resulting schedule is

$$\bar{\mathbf{t}} = (0, 63.889, 87.557, 109.557, 132.538, 154.538, 218.210, 240.380, 263.828, 286) \mu\text{s}. \quad (26)$$

The largest boundary correction is $2.111 \mu\text{s}$, and the infinity-norm closure error is $1.421\text{e-}14$. We use this reconciliation only to prepare the benchmark data; it is not a trajectory-design method.

The cycle averages in [figure 1](#) are 10.41056 V , 19.42228 V , and 0.95613 A . We design the controller in the normalized coordinates $\mathbf{z} = S_x^{-1}\mathbf{e}$ and $\mathbf{v} = \delta\boldsymbol{\tau}/t_s$. The aggressive raw law is a standard discrete LQR with $Q_n = I_3$ and $R_n = 0.001I_8$. The conservative comparison law uses $R_n = 0.1I_8$. We condition only the aggressive law.

4.2 Linearization and certificate checks

We check the analytical Jacobians against central differences of the exact one-cycle map. The state steps are 10^{-4} times the diagonal entries of S_x , and the timing step is $10^{-3}t_s = 0.01 \mu\text{s}$. A second check applies simultaneous state and timing perturbations with normalized amplitude ϵ and

evaluates

$$r(\epsilon) = \left\| S_x^{-1} [\mathcal{P}(\mathbf{x}^* + \Delta\mathbf{x}, \delta\boldsymbol{\tau}) - \mathbf{x}^* - \Phi\Delta\mathbf{x} - \Gamma_\tau\delta\boldsymbol{\tau}] \right\|_2. \quad (27)$$

The Jacobian errors and residual slope in [table 2](#) check the implemented first-order model. The generated numerical data include the full analytical and finite-difference matrices and the residual sweep.

For the aggressive normalized closed loop, an offline semidefinite program produces a common P . The verified endpoint norms are $\|\Phi\|_P = 0.999939$ and $\|A_{\text{cl}}\|_P = 0.979743$, which give the segment bound 0.999939 in [proposition 2](#). This bound applies to the conditioned linearized family. The controller solves no semidefinite program online.

4.3 Large-error simulation

We simulate the exact nonlinear map for 100 cycles from

$$\mathbf{x}_0 = [7.5143 \quad 20.8211 \quad 0.0314]^\top. \quad (28)$$

We compare the conditioned aggressive law, the unconditioned conservative law, and open loop from the same initial state. Exact matrix exponentials propagate every interval; the simulation does not use the linearized model. The initial normalized error is 0.983946 , well outside the perturbation range used to check the local linearization.

Without conditioning, the aggressive law requests a minimum dwell duration of $-43.045 \mu\text{s}$, which would invert an interval. The conditioner acts for 3 cycles and reaches $\beta_{\min} = 0.293978$. It reduces the maximum offset from 65.217 to $19.172 \mu\text{s}$. The minimum applied dwell is $3.000 \mu\text{s}$, equal to the assumed bound. The conservative raw law remains feasible and has a minimum dwell of $15.502 \mu\text{s}$. It crosses the 0.01 error threshold at cycle 20, compared with cycle 6 for the conditioned aggressive law.

The 100-cycle state-space view alone can make the open-loop response appear divergent because its contraction is very slow. Under the fixed nominal schedule, however, the open-loop cycle map has spectral radius $\rho(\Phi) = 0.999850 < 1$, so the cycle-start error converges asymptotically to zero. It

Table 1 Converter, schedule, and feedback parameters. The applied bound of $3\text{ }\mu\text{s}$ is a simulation assumption, not the nominal benchmark-design bound.

Quantity	Symbol	Value
Source; capacitances	$E; C_1, C_2$	30 V; 40 μF
Inductance; resistance	$L; R$	10 mH; 10 Ω
Intervals; period	$N; T$	9; 286 μs
Nominal-design dwell bound	d_{bench}	22 μs
Applied-schedule dwell bound	d_{min}	3 μs
State; timing scale	$S_x; t_s$	diag(10, 20, 1); 10 μs
Aggressive LQR weights	$Q_n; R_n$	$I_3; 0.001I_8$
Conservative LQR weights	$Q_n; R_n$	$I_3; 0.1I_8$

Table 2 Linearization checks, stability certificate, and exact nonlinear simulation results. Settling denotes the first cycle for which $\|S_x^{-1}e_k\|_2 < 0.01$.

Quantity	Value
Relative errors: $\Phi; \Gamma_\tau$	8.518e-12; 8.344e-11
Linearization residual slope	2.000
Spectral radii: open; conservative; aggressive	0.999850; 0.941468; 0.568151
Certified conditioned-family contraction bound	0.999939
Minimum conditioning factor; active cycles	0.293978; 3
Maximum raw; applied offset (μs)	65.217; 19.172
Minimum raw; applied dwell (μs)	-43.045; 3.000
Minimum conservative dwell (μs)	15.502
Settling cycle: conditioned; conservative	6; 20
Final errors: conditioned; conservative; open	1.746e-15; 3.113e-06; 0.298010

remains above the normalized-error threshold 0.01 until cycle 22009, whereas the conditioned response crosses that threshold at cycle 6. Figure [figure 4](#) shows both trajectories over the initial transient and the subsequent long-horizon convergence.

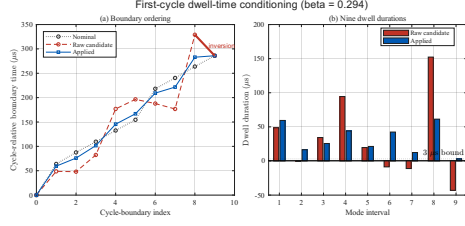


Figure 2 Dwell-time conditioning during the first cycle. (a) Nominal, raw, and applied cycle boundaries. The highlighted descending segment marks a boundary inversion in the raw schedule; conditioning preserves the boundary order. (b) Raw and applied durations for all nine dwells. The raw duration reaches $-43.045 \mu\text{s}$, while the applied schedule enforces the assumed $3 \mu\text{s}$ bound.

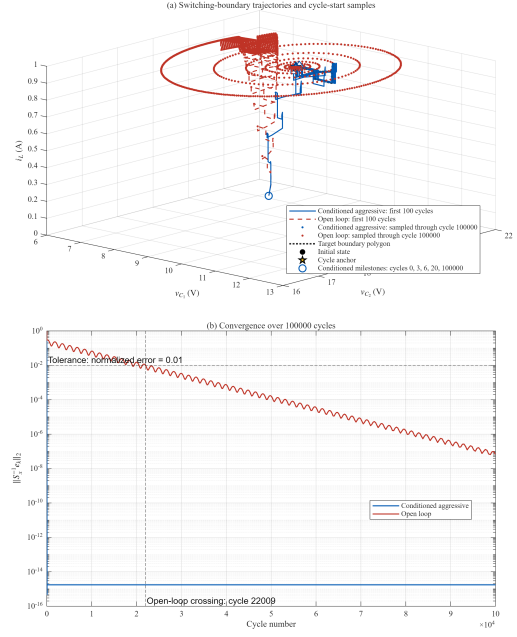


Figure 4 Exact large-error trajectory comparison over 100000 cycles. (a) The first 100 cycles are shown as exact switching-boundary paths; later points are cycle-start samples through cycle 100000 for both conditioned aggressive feedback and open loop. Markers identify conditioned cycle starts at cycles 0, 3, 6, 20, and 100000. (b) Normalized cycle-start errors over the common horizon. The open-loop response converges to the target switching limit cycle but remains above 0.01 until cycle 22009, whereas the conditioned response crosses it at cycle 6. These simulations are benchmark evidence and do not establish global nonlinear stability.

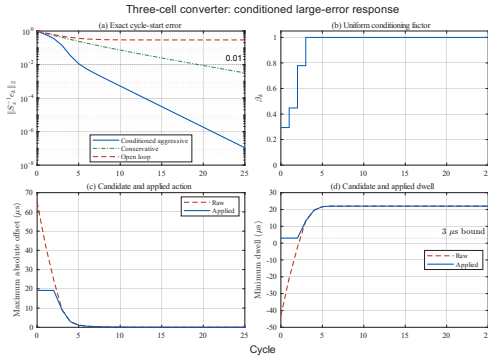


Figure 3 Exact nonlinear response over 100 cycles from the large-error initial state. (a) Normalized cycle-start errors with conditioned aggressive feedback, unconditioned conservative feedback, and open loop. (b) Conditioning factor. (c) Maximum absolute raw and applied switching-instant offsets. (d) Minimum raw and applied dwell durations. The raw duration becomes negative, while the applied schedule satisfies the assumed $3 \mu\text{s}$ bound.

5 Discussion

The conditioner retains the largest feasible scalar multiple of a static timing action without changing its direction or the final cycle boundary. It does not replace constrained optimization when the controller must predict future cycles or change the action direction. The search for a common P runs offline and adds no online computation.

The model combines results from earlier work. Patiño et al. derived the mode-dependent switching sensitivity Patiño et al. (2010). Marcolino et al. formulated fixed-period coordinates, dwell constraints, and LQR for switched actuators Marcolino et al. (2021). Our derivation connects these formulations and checks the resulting model numerically. The new control result is the scalar conditioner and its common-Lyapunov certificate, not the LQR design.

The 3 μ s minimum dwell is a simulation assumption. The study omits timer quantization, delays, dead time, hardware margins, state estimation, model uncertainty, and experiments. It also fixes the mode order, phase, and period. The exact nonlinear stability result holds only in a neighborhood where conditioning is inactive. Proposition 2 applies to the linearized family, so the large-error simulation is evidence for this benchmark rather than a global nonlinear guarantee.

6 Conclusion

The proposed scalar conditioner enforces every minimum-dwell constraint in $O(N)$ time while preserving the action direction and cycle period. A common quadratic Lyapunov function for the endpoint matrices certifies the full conditioned linearized family. The augmented derivation covers mode-dependent switched-affine dynamics and reduces to the switched-actuator model. Numerical differentiation checks the resulting Jacobians. In exact nonlinear simulation, conditioning makes an infeasible aggressive action admissible, and the resulting response settles faster than a feasible conservative law. Hardware tests remain necessary to evaluate timer quantization, delays, and implementation margins. The nonlinear stability guarantee remains local.

Declarations

Draft TODO before submission:

- Add the Funding statement.
- Add the Competing Interests statement.
- Add the Data Availability statement, which is required for original research.
- Add the Code Availability statement, or state that it is not applicable.
- Add the Author Contributions statement.
- Add an Acknowledgements section if applicable.
- Add the author ORCID if available.

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